

Project Report

Group 39: Bahar Abit, Hüseyin Eren Yıldız, İpek Uzun, Neşenaz Yalçın, Sait Kaçmaz

1. Introduction

The goal of this project is to solve Task 1 of the SemEval-2026 CLARITY challenge. This task focuses on political interviews and asks how clearly a politician answers a question. In politics, speakers often dodge questions or respond vaguely on purpose. Our system takes a question–answer pair and assigns one of three labels: Clear Reply, Ambivalent, or Clear Non-Reply. This is challenging because political language is nuanced, and the boundary between being unclear and refusing to answer can be subtle.

We first conducted a literature review, followed by exploratory data analysis and preprocessing on the QEvason dataset. Each instance was represented as a QA pair and used for supervised learning. As baselines, we trained TF-IDF models including Logistic Regression, Linear SVM, and Decision Trees, and we also implemented XGBoost on TF-IDF features as a stronger non-linear baseline. We report macro-averaged F1 as the primary metric to measure balanced performance across the three classes.

Our best traditional machine learning baseline achieved a macro F1-score of 0.59. To better capture the nuances of political evasion, we fine-tuned transformer models and compared DeBERTa-v3-small and DeBERTa-v3-base. The DeBERTa-v3-base model achieved the best overall performance with a macro F1-score of 0.70, and it notably improved detection of Clear Non-Reply instances.

2. Related Work

Recent NLP work frames response clarity as a supervised question–answer pair classification problem, which matches CLARITY Task 6 (Subtask 1): given a question and its answer, predict whether the answer is a clear reply, an ambivalent reply, or a clear non-reply. A key reference is Thomas et al. (2024), which introduces a taxonomy for response clarity/evasion and provides dataset construction and baseline modeling choices that closely mirror CLARITY’s formulation.

From the modeling side, modern approaches typically fine-tune pretrained transformer encoders on sentence-pair inputs. Devlin et al. (2019) show that BERT-style encoders can be fine-tuned effectively for pairwise classification tasks by jointly encoding two text segments, which directly supports treating (question, answer) as the input and predicting a discrete clarity label. For capturing question–answer alignment beyond raw lexical overlap, Reimers and

Gurevych (2019) propose Sentence-BERT, enabling efficient semantic similarity scoring; such embedding-based similarity is commonly used as a signal for whether an answer addresses the question content, which is conceptually tied to clarity.

Clarity classification is also related to work on identifying when answers are not truly responsive or supported. In extractive QA, Rajpurkar et al. (2018) introduce unanswerable questions (SQuAD 2.0), motivating explicit detection of “no-answer” cases rather than forcing a model to produce an answer; while CLARITY differs from extractive QA, the underlying idea of recognizing non-responsiveness (or inability to answer) is closely aligned with separating clear replies from clear non-replies. Finally, since political interviews are multi-turn and context-dependent, research on conversational QA helps motivate robust QA-pair handling in dialogue-like settings. Choi et al. (2018) (QuAC) formalize question answering in an information-seeking conversational setup and highlight challenges such as context carryover and partial answers, which resemble the ambiguity and indirectness seen in interview responses.

Overall, these works motivate the standard CLARITY-style pipeline used in our project: represent each instance as a QA pair, learn a classifier with transformer encoders (BERT-family), and evaluate with class-balanced metrics (macro-F1), while also considering semantic alignment signals that correlate with clarity.

3. Methodology

We utilized the official CLARITY dataset, which consists of QA pairs from political interviews. The dataset includes question–answer pairs, from U.S. Press briefings, with each answer marked by a clarity label that reflects how directly the politician responded to the journalist’s inquiry. For Subtask 1, the dataset offers three classification labels: Clear Reply, Clear Non-Reply, and Ambivalent. The dataset comprises 3,760 entries in total, split into a training set of about 3,450 examples and a test set of approximately 308 examples. All information is provided in Parquet format. Contains extra metadata fields including speaker, date, event and detailed evasion classifications.

We loaded the QEvasion/CLARITY dataset directly from Hugging Face in Parquet format and constructed our modeling input by pairing the extracted sub-question (question) with the corresponding response (interview_answer). We kept preprocessing minimal so we wouldn’t remove useful signals: we applied Unicode normalization (NFKC), removed leading/trailing whitespace, and collapsed repeated whitespace. For exploratory data analysis, we computed word level length statistics for questions and answers and examined their distributions across the three clarity labels. In the official training set, label frequencies were Ambivalent (2040), Clear Reply (1052), and Clear Non-Reply (356), indicating a notable class imbalance. We visualized label distribution and answer-length distributions by label, and we report these figures in the Appendix.

Since the official release provides train and test partitions, we created an internal validation set by stratified splitting of the training data into 85% train and 15% validation using a fixed random seed (SEED = 42) to preserve label proportions. To address the imbalance, we computed class weights from the training partition only (inverse frequency weighting) and used them in a weighted cross entropy loss during training. This ensures that minority class errors contribute proportionally more to the optimization objective while leaving validation and test distributions unchanged.

Additionally, we implemented a DeBERTa-v3-base (Decoding-enhanced BERT with disentangled attention) as proposed by He et al. (2020). While standard models like BERT and RoBERTa are common benchmarks, we chose DeBERTa-v3 specifically for its superior handling of semantic nuance in Question-Answering pairs. Unlike BERT, which represents each word as a single vector, DeBERTa represents words using two separate vectors. This allows the model to better understand the relative relationship between a politician's response and the initial question. Specifically, we used 10 epochs with a learning rate of 3×10^{-5} , and the sequence length was 300.

Moreover, our training pipeline was optimized for performance. We used a weighted trainer to override the loss function. This allows the integration of our pre-calculated weights_tensor directly into the loss calculation. To simulate a larger, more stable batch size of 32 while fitting within VRAM limits, we used a train batch_size of 8 with gradient accumulation steps set to 4. Finally, we enabled 16-bit floating-point training to accelerate computation and reduce the memory footprint.

4. Results

4.1 Experimental Setup and Metrics

All experiments were evaluated on a held-out validation split of 518 QA pairs with the following class distribution: Ambivalent (306), Clear Non-Reply (54), and Clear Reply (158). We report macro-F1, macro precision, and macro recall as the primary metrics to reflect balanced performance across classes. In addition, we present confusion matrices and one-vs-rest precision-recall (PR) curves to analyze class-level behavior.

4.2 Overall Performance Comparison

Table 1 shows a clear performance gap between classical feature-based models and transformer fine-tuning. The strongest classical baseline, TF-IDF + Logistic Regression (lr_unigram), achieves macro-F1 = 0.590, while TF-IDF + XGBoost offers only a modest improvement (0.601). In contrast, fine-tuning DeBERTa-v3-base yields a large jump to macro-F1 = 0.712, suggesting that contextual encoders capture clarity cues that sparse n-gram features miss. Classical models tend to have relatively higher macro precision but lower macro recall,

indicating difficulty in retrieving borderline cases; DeBERTa improves macro recall (0.743) and overall macro-F1, consistent with modeling longer context and pragmatic signals (e.g., indirectness, partial relevance).

	model	macro_f1	macro_precision	macro_recall	amb_label	amb_true_count	amb_error_count	amb_error_rate
0	deberta_v3_base	0.712325	0.692508	0.742547	Ambivalent	306	87	0.284314
1	xgb_tfidf	0.600968	0.609059	0.595203	Ambivalent	306	75	0.245098
2	tfidf_best	0.590133	0.587459	0.593061	Ambivalent	306	85	0.277778
3	deberta_v3_small	0.587907	0.569343	0.618235	Ambivalent	306	105	0.343137
4	fasttext_xgb	0.574837	0.615156	0.554411	Ambivalent	306	57	0.186275

Table 1. Metric Comparison

Within TF-IDF baselines, Logistic Regression outperforms Linear SVM and Decision Tree, implying a linear decision boundary is sufficient for the strongest lexical cues. FastText + XGBoost underperforms TF-IDF + XGBoost, likely because mean-pooled static embeddings lose discriminative, question-conditioned lexical triggers. Finally, we focus on macro-F1 because it weights all three labels equally and better reflects performance under ambiguity, especially for the Ambivalent class.

4.3 Analysis of models

For our results, we found that our model is quite good at spotting clear answers and evasive ones, reaching an overall accuracy of 70% using DeBERTa-v3-base. Looking at the Confusion Matrix, we can see the model is best at identifying "Ambivalent" responses, correctly catching 204 of them. However, it sometimes gets confused between "Ambivalent" and "Clear Reply," which makes sense because the difference between a direct answer and a slightly vague one can be very small. The Precision-Recall curves show that our model stays fairly steady, especially for the "Ambivalent" category (with an AP of 0.84), while identifying "Clear Non-Replies" is a bit more of a challenge. Even so, our system did better than the basic models mentioned in the original research papers, mostly because we used special settings to make sure the model didn't ignore the smaller groups of data.

Our journey to improve the system was a process of trial and error that led us from a starting score of 0.59 to a final 0.70 Macro F1-score. We began with a smaller, faster model (DeBERTa-v3-small), but we realized it didn't have enough power to understand the tricky ways politicians dodge questions. By switching to the DeBERTa-v3-base model and letting it read more text at once, the model started picking up on subtle clues it had missed before. We also gave the model more time to learn by increasing the training rounds from 4 to 10 and used Gradient Accumulation. This acted like a larger "study group" for the model, making its learning much more stable and reliable. These changes, combined with a slightly faster learning speed,

allowed our system to better handle the complex and often confusing language found in real-world political interviews.

In addition to the transformer experiments, we analyzed our TF-IDF baselines to understand what purely lexical features capture in this task. Among these models, TF-IDF + Logistic Regression (lr_unigram) performed best (macro-F1 = 0.590), while TF-IDF + Decision Tree (dt_unigram) and TF-IDF + Linear SVM (svm_unigram) lagged behind (macro-F1 = 0.548 and 0.542, respectively). This gap suggests that simple linear weighting of n-gram features is a strong lexical baseline, whereas more complex decision boundaries (trees) or margin-based training (SVM) do not provide additional benefit under the same sparse representation. These TF-IDF results serve as a reference point for interpreting the gains obtained by XGBoost and DeBERTa.

Additionally, two baseline models were tested using XGBoost with different feature representations. The first model uses TF-IDF features and achieves an accuracy of 0.65, a macro-F1 score of 0.60, and a weighted-F1 score of 0.65. Class-wise F1-scores are 0.73 for Ambivalent, 0.55 for Clear Non-Reply, and 0.53 for Clear Reply. The confusion matrix shows that most Ambivalent samples are correctly classified, while a significant number of Clear Reply samples are predicted as Ambivalent. Precision–recall curves computed in a one-vs-rest setting show an average precision of 0.75 for Ambivalent, 0.54 for Clear Non-Reply, and 0.55 for Clear Reply.

The second model combines XGBoost with FastText mean word embeddings of 200 dimensions. This model achieves an accuracy of 0.65, a macro-F1 score of 0.57, and a weighted-F1 score of 0.64. Class-wise F1-scores are 0.74 for Ambivalent, 0.51 for Clear Non-Reply, and 0.47 for Clear Reply. Compared to the TF-IDF model, confusion between Clear Reply and Ambivalent increases. Precision–recall curves show average precision values of 0.73 for Ambivalent and approximately 0.52 for both Clear Non-Reply and Clear Reply.

Overall, both baseline models achieve similar accuracy and macro-F1 scores. Performance is consistently higher for the Ambivalent class, while distinguishing between Clear Reply and Clear Non-Reply remains challenging. These results are lower than transformer-based approaches reported in the QEvasion literature, which is expected given the use of classical feature representations and tree-based models. The reported experiments provide baseline results and inform the selection of more advanced models in later stages of the project.

4.4 Our Journey: What We Tried

Our development followed an iterative process:

Step 1: TF-IDF baselines. We started with TF-IDF representations and trained Logistic Regression, Linear SVM, and Decision Tree models, tuning feature setups (e.g., n-grams) and model hyperparameters. The best TF-IDF baseline was `lr_unigram` with $\text{macro-F1} = 0.590$.

Step 2: XGBoost baselines. We trained XGBoost with two feature representations (TF-IDF and FastText mean) to test whether tree-based ensembles improve over linear baselines. TF-IDF + XGBoost delivered a modest improvement ($\text{macro-F1} = 0.601$).

Step 3: Transformer fine-tuning. We then fine-tuned DeBERTa-v3-small and DeBERTa-v3-base. The base model provided a large performance jump to $\text{macro-F1} = 0.712$, and became our final model. We also improved training stability by increasing training duration (e.g., up to 10 epochs) and using gradient accumulation, enabling more reliable optimization under limited batch sizes.

5. Discussion

The choice of dataset has a direct impact on the performance of the proposed system. We used the QEvation dataset, which consists of real political interview question–answer pairs annotated for response clarity. This dataset reflects authentic political language, where answers are often long, indirect, and deliberately ambiguous. As a result, the task is inherently difficult, particularly for distinguishing between Clear Reply and Ambivalent responses. In addition, the dataset is highly imbalanced, with Ambivalent being the dominant class and Clear Non-Reply representing a small minority. This imbalance is clearly reflected in the Results section, where all evaluated models achieve their strongest performance on Ambivalent, while Clear Non-Reply remains the most challenging label. Although class-weighted training helps alleviate this issue, the dataset composition still limits minority-class performance.

Our choice of modeling approach balances simplicity, interpretability, and representational power. Classical models based on TF-IDF and FastText features provide efficient and transparent baselines, achieving macro F1-scores of up to 0.60. Their main advantage lies in fast training and ease of analysis; however, these models rely largely on surface-level lexical patterns and therefore struggle to capture subtle semantic relationships between questions and answers. This limitation is evident in their confusion between Clear Reply and Ambivalent responses. In contrast, the transformer-based DeBERTa-v3-base model offers a richer contextual representation and significantly improves performance, reaching a macro F1-score of 0.70. The main disadvantage of this approach is its higher computational cost and sensitivity to hyperparameters such as sequence length, number of epochs, and batch size configuration.

When comparing our results to existing systems in the literature, our findings are consistent with prior work on response clarity and evasiveness detection. Previous studies, including Thomas et al. (2024), report that classical feature-based approaches show limited effectiveness in separating Ambivalent from Clear Reply classes, which matches the behavior

observed in our baseline models. Transformer-based models are reported to achieve higher macro-level performance by better modeling question–answer alignment, a trend that is also confirmed by our DeBERTa-v3-base results. While some studies report higher absolute scores using larger or domain-adapted models, our results remain competitive given the use of a base-sized transformer and a standard fine-tuning setup.

Despite the improvements achieved, the proposed system has several limitations. First, each question–answer pair is processed independently, without explicitly modeling broader conversational context or speaker history, which may be relevant for interpreting evasive behavior. Second, the system produces discrete class labels, even though ambiguity often exists on a continuum rather than as clearly separable categories. Finally, evaluation focuses on quantitative metrics and does not fully capture qualitative differences in evasive strategies across classes.

If more time and resources were available, several improvements could be explored. Incorporating discourse-level or multi-turn context could help reduce confusion between Clear Reply and Ambivalent. Multi-task learning that jointly predicts clarity labels and fine-grained evasion types could further improve robustness. In addition, experimenting with larger or domain-adapted transformer models, as well as targeted data augmentation for the Clear Non-Reply class, may lead to further performance gains. More detailed qualitative error analysis could also guide the development of features or training strategies tailored to specific evasive patterns.

6. Conclusion

In this project, we addressed Subtask 1 of the SemEval-2026 CLARITY challenge: given a political interview question and its answer, predict whether the answer is a Clear Reply, Ambivalent, or Clear Non-Reply. This task is challenging because political responses often contain subtle forms of evasion, and the classes are imbalanced, especially for Clear Non-Reply.

We built a full pipeline from data loading and lightweight preprocessing to model training and evaluation. Our classical baselines using TF-IDF features (including logistic regression/SVM-style models and XGBoost variants) achieved around 0.59 Macro F1, and showed consistent confusion between Clear Reply and Ambivalent. Our best model was a DeBERTa-v3-base transformer fine-tuned on question/answer pairs, which improved performance to 0.70 Macro F1. Importantly, this model increased our effectiveness on the minority Clear Non-Reply class, where the F1-score improved substantially compared to the baseline. Using a stratified validation split and class weighted cross entropy helped ensure the model did not ignore the minority class and led to more balanced performance across labels.

Overall, our results suggest that transformer based question/answer pair encoders are better suited for capturing the nuanced relationship between a question and a political response than classical feature based approaches.

7. Individual Contributions

Neşenaz managed the data pipeline, including loading the QEvasion dataset, cleaning/normalizing text, performing EDA (label and length distributions), and preparing the train/validation splits. She also coordinated efforts to address class imbalance (e.g., class weights/resampling).

Sait implemented and tuned TF-IDF baselines (Logistic Regression, SVM, Decision Trees), experimenting with different TF-IDF settings (e.g., n-grams) and performing cross-validation. İpek and Neşenaz supported validation of these results.

İpek led the transformer track, fine-tuning pretrained encoders (e.g., DeBERTa-style models) for QA-pair classification, managing training/hyperparameters, and comparing transformers against classical baselines.

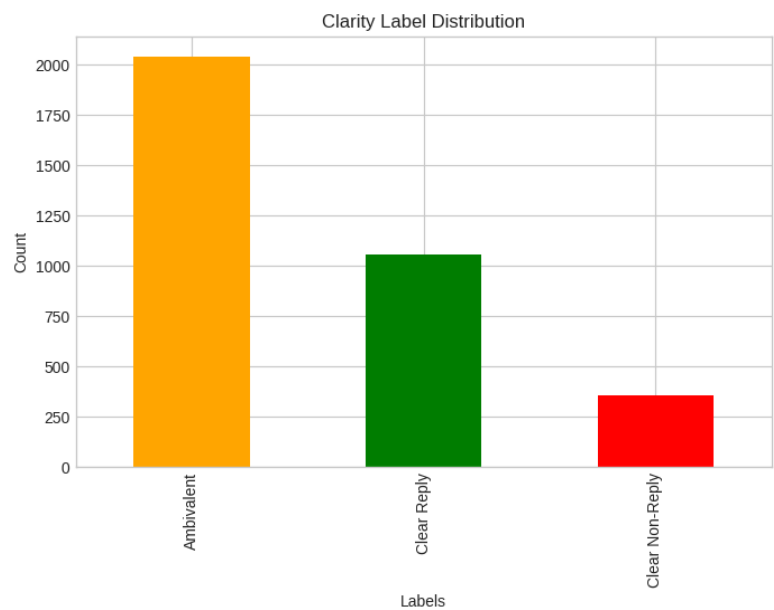
Bahar implemented and tuned the multi-class XGBoost model with TF-IDF features, handled imbalance via per-sample weights, and integrated probability outputs into the shared evaluation pipeline (metrics, confusion matrix, PR curves). She also supported qualitative error analysis, especially for the Ambivalent class.

Eren built the evaluation and reporting pipeline (macro-F1/precision/recall, confusion matrices, PR curves) and conducted error analysis, focusing on Ambivalent; he also assisted Neşenaz with data checks.

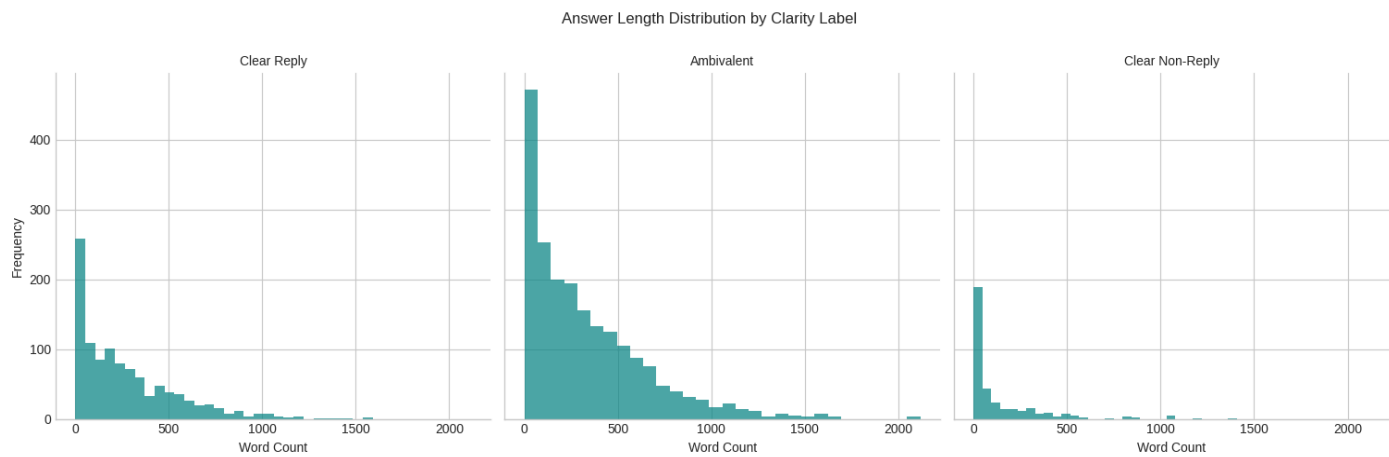
Finally, all team members collaborated on updating the literature review, writing the final report, and preparing presentation materials.

APPENDICES

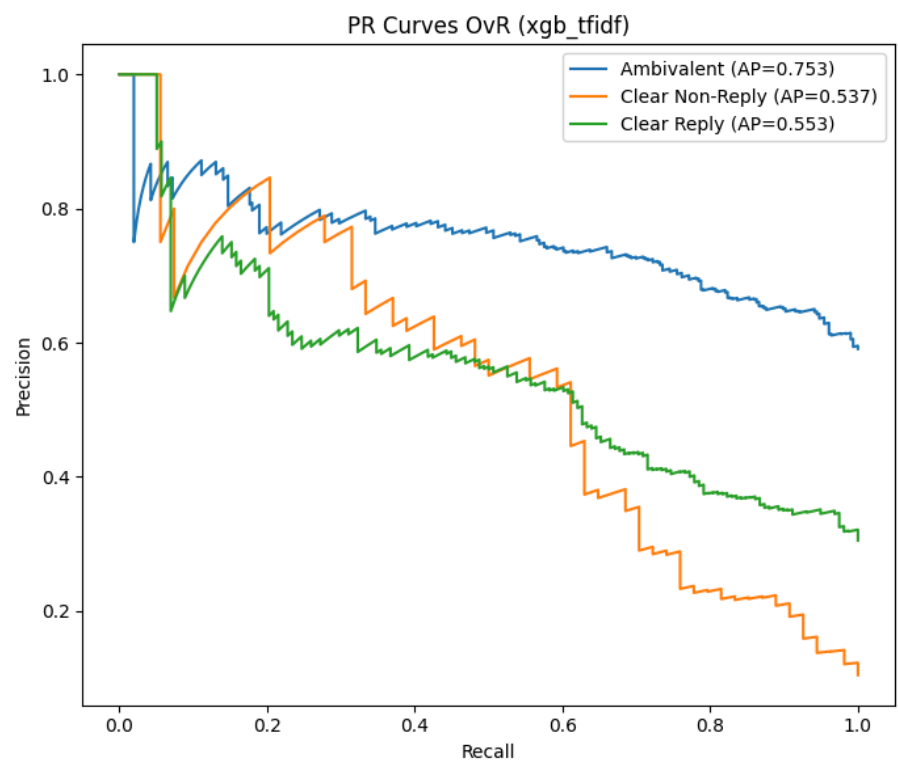
Appendix A: Dataset Overview and Label Distribution



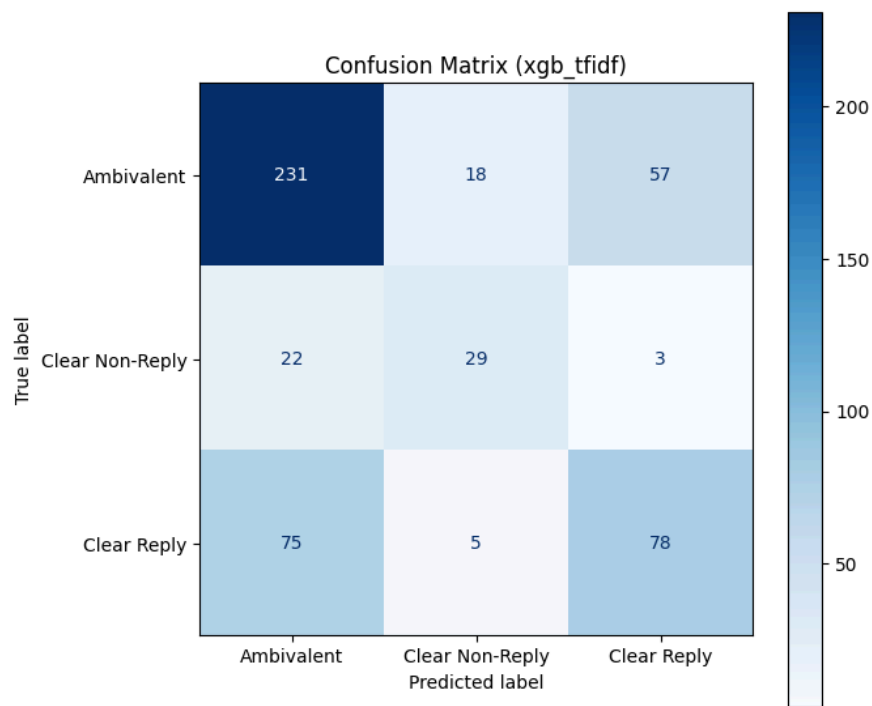
Appendix B: Text Length Analysis



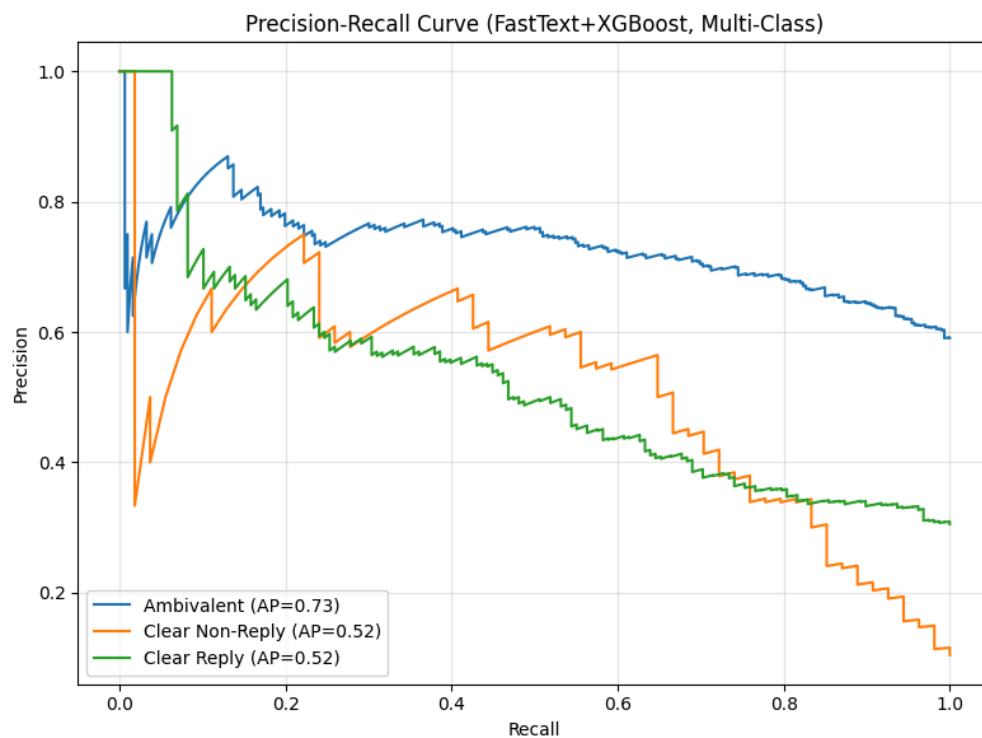
Appendix C: Precision Recall Curve for XGBoost (TF-IDF)



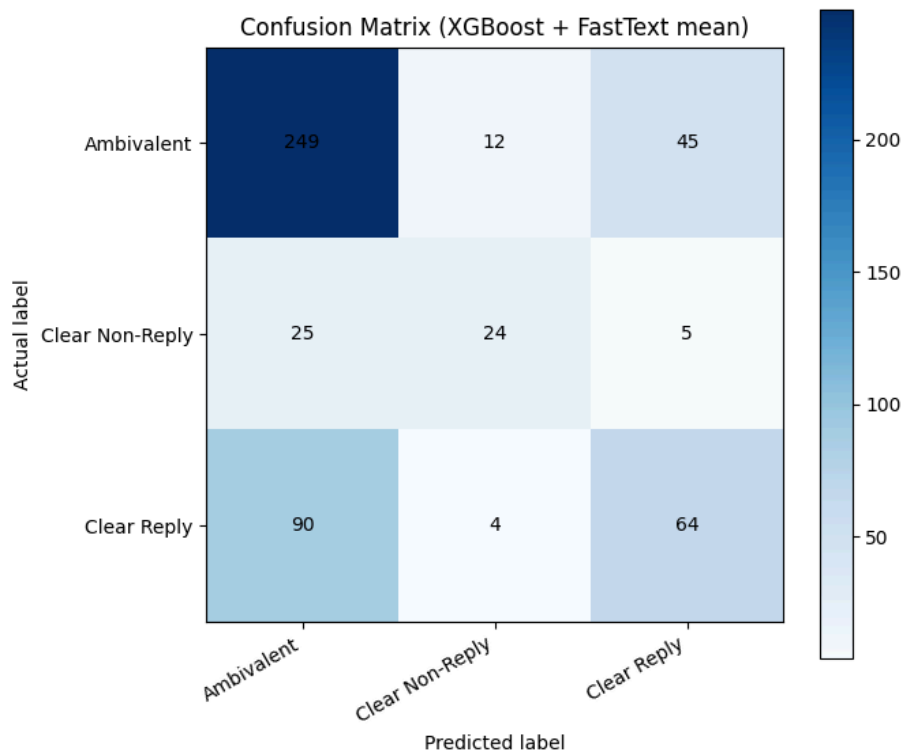
Appendix D: Confusion Matrix for XGBoost (TF-IDF)



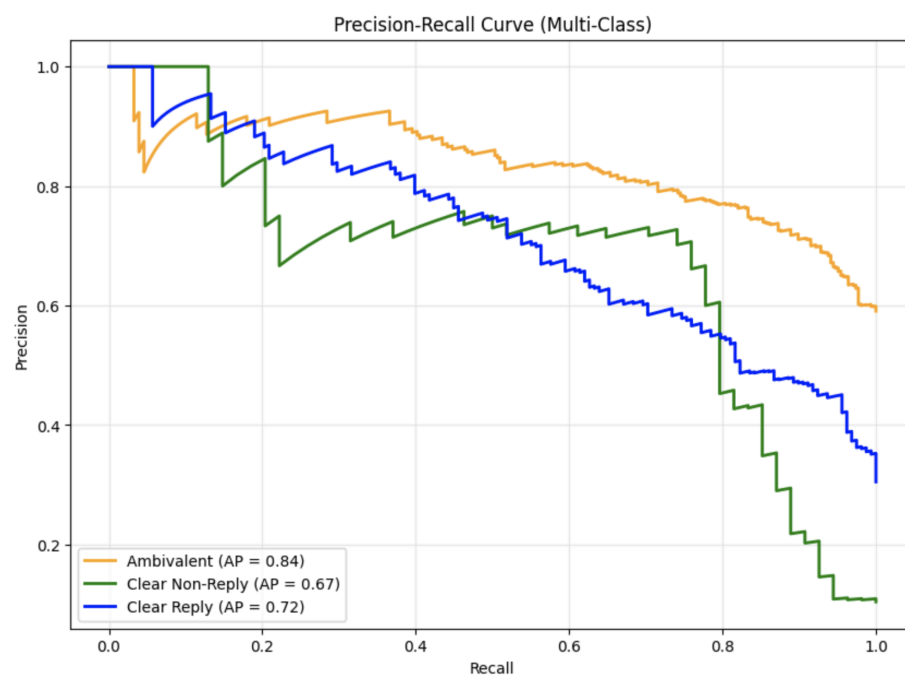
Appendix E: Precision Recall Curve for XGBoost (FastText)



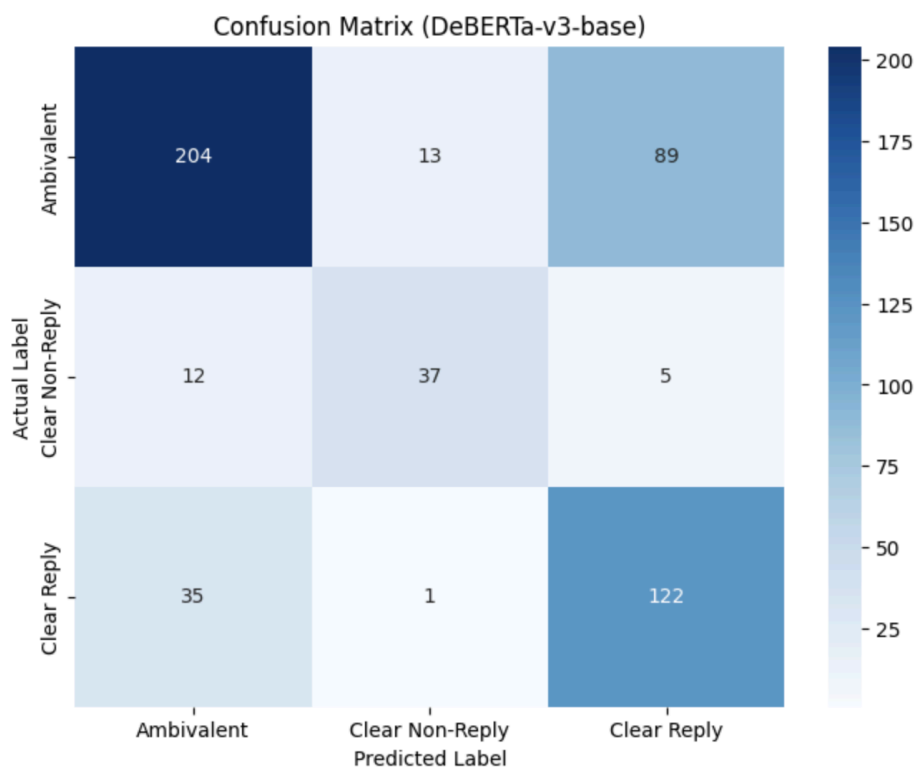
Appendix F: Confusion Matrix for XGBoost (FastText)



Appendix G: Precision Recall Curve for DeBERTa-v3-base



Appendix H: Confusion matrix for DeBERTa-v3-base



Appendix I: Proof of Submission

ID # ▾	File name	Date	Status	Score	Actions
476848	prediction.zip	2026-01-05 21:11	Finished	0.58	  
476840	submission.csv.zip	2026-01-05 21:00	Failed 		

References

- He, P., Liu, X., Gao, J., & Chen, W. (2020). DeBERTa: Decoding-enhanced BERT with Disentangled Attention. *ArXiv*. <https://arxiv.org/abs/2006.03654>
- He, P., Gao, J., & Chen, W. (2021). DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. *ArXiv*. <https://arxiv.org/abs/2111.09543>
- Thomas, K., Filandrianos, G., Lymperaious, M., Zerva, C., & Stamou, G. (2024). “I never said that”: A dataset, taxonomy and baselines on response clarity classification. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 5204–5233. <https://doi.org/10.18653/v1/2024.findings-emnlp.300>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 4171–4186). Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423> aclanthology.org
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 3982–3992). Association for Computational Linguistics. <https://doi.org/10.18653/v1/D19-1410> aclanthology.org
- Rajpurkar, P., Jia, R., & Liang, P. (2018). Know what you don't know: Unanswerable questions for SQuAD. In I. Gurevych & Y. Miyao (Eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (pp. 784–789). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P18-2124> aclanthology.org
- Choi, E., He, H., Iyyer, M., Yatskar, M., Yih, W.-t., Choi, Y., Liang, P., & Zettlemoyer, L. (2018). QuAC: Question answering in context. In E. Riloff, D. Chiang, J. Hockenmaier, & J. Tsujii (Eds.), *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*

(pp. 2174–2184). Association for Computational Linguistics.

<https://doi.org/10.18653/v1/D18-1241>